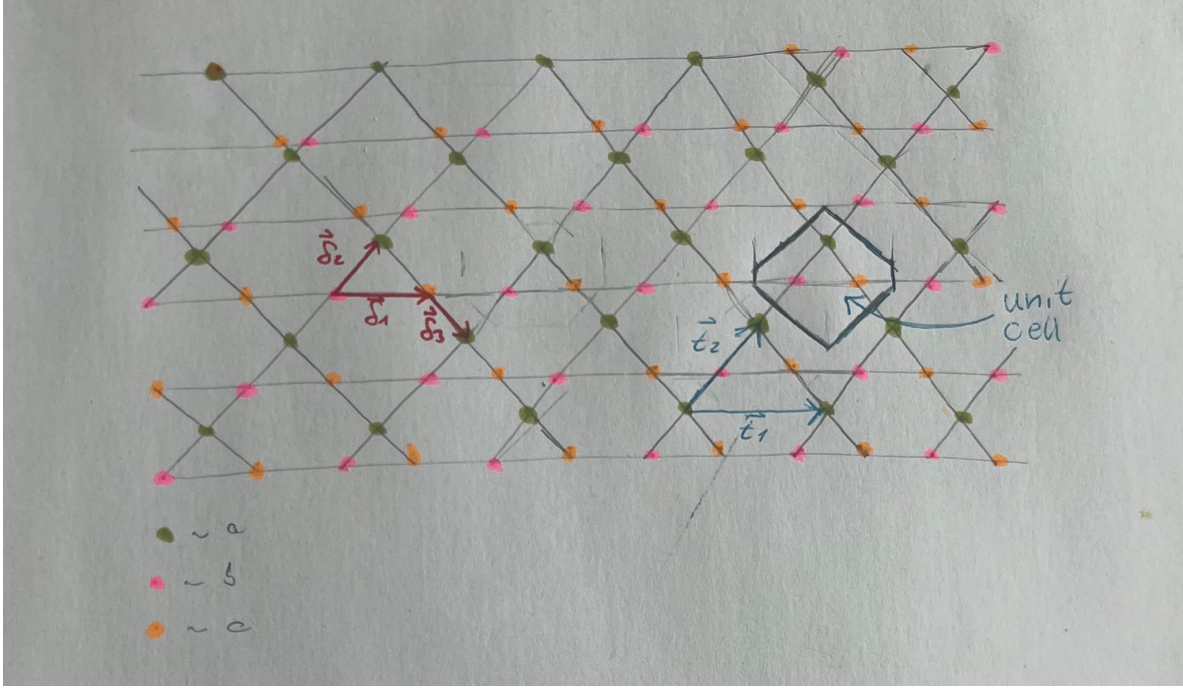


0.1 0-flux State

0.1.1 Real Space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise ;)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \quad (1)$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \quad (2)$$

The tight binding hamiltonian is

$$H = t \sum_{i\sigma} \left(a_{i\sigma}^\dagger c_{i\sigma} + c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + c_{i\sigma}^\dagger b_{i+t_1;\sigma} + c_{i\sigma}^\dagger a_{i+t_1-t_2;\sigma} + a_{i\sigma}^\dagger b_{i+t_2;\sigma} + \text{h.c.} \right) \quad (3)$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

0.1.2 Reciprocal Space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix} \quad (4)$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \sqrt{\frac{3}{N}} \sum_k e^{i\vec{k}\cdot\vec{R}_i} a_{k\sigma}, \quad (5)$$

where N is the number of unit cells. This yields

$$H = 2t \sum_{q\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger & c_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & \cos(\vec{q} \cdot \vec{\delta}_2) & \cos(\vec{q} \cdot \vec{\delta}_3) \\ \cos(\vec{q} \cdot \vec{\delta}_2) & 0 & \cos(\vec{q} \cdot \vec{\delta}_1) \\ \cos(\vec{q} \cdot \vec{\delta}_3) & \cos(\vec{q} \cdot \vec{\delta}_1) & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix}. \quad (6)$$

0.1.3 dispersion and band touching points

The three eigenvalues are

$$\varepsilon_1(\vec{q}) = t \left(1 - \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (7)$$

$$\varepsilon_2(\vec{q}) = t \left(1 + \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (8)$$

$$\varepsilon_3(\vec{q}) = -2t, \quad (9)$$

see [1]. I also used

$$\cos^2\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos^2\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) = \cos(q_x) \cos(\sqrt{3}q_y) + 1 \quad (10)$$

Interestingly, the top band is completely flat. The other two bands touch when

$$\cos^2(q_x) + \cos(q_x) \cos(\sqrt{3}q_y) + 1/4 = 0 \quad (11)$$

$$\Rightarrow \cos(q_x) = \frac{1}{2} \left(-\cos(\sqrt{3}q_y) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1} \right) \quad (12)$$

$$\Rightarrow q_y = \frac{\pi l}{\sqrt{3}}, \quad l \in \mathbb{N} \quad (13)$$

$$\Rightarrow q_x = (-1)^s \left(\frac{\pi}{2} + \frac{(-1)^l \pi}{6} \right) + 2\pi m, \quad m \in \mathbb{N}, \quad s \in [0, 1] \quad (14)$$

These touching points lie at the corners of the Brillouin zone, there are two "species". One can imagine that two of the parameters represent translation by reciprocal lattice vectors while the third one selects the species of the touching points. This not what m , l and s do but maybe one can transform them. Also, why does the solution depend on three parameters? For example

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad m = 0, s = 0, l = 0 \quad (15)$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad m = 0, s = 1, l = 0. \quad (16)$$

0.1.4 Filling Factor

The idea is the following: The lower two bands contain the same number of states by symmetry. By calculating the number of states of the upper band (should be easy) it should be possible to calculate at what filling factor the low energy physics happen exactly the linear touching points.

0.1.5 Linearization in Reciprocal Space and Projection

Use

$$\cos\left(\left(\vec{K}^{(\prime)} + \vec{p}\right) \cdot \vec{\delta}_{ij}\right) \approx \cos\left(\vec{K}^{(\prime)} \cdot \vec{\delta}_{ij}\right) - \sin\left(\vec{K} \cdot \vec{\delta}_{ij}\right) \left(\vec{p} \cdot \vec{\delta}_{ij}\right) \quad (17)$$

and

$$\begin{aligned} \vec{K} \cdot \vec{\delta}_1 &= \frac{2\pi}{3} \\ \vec{K} \cdot \vec{\delta}_2 &= \frac{\pi}{3} \\ \vec{K} \cdot \vec{\delta}_3 &= \frac{\pi}{3} \\ \vec{K}' &= -\vec{K} \end{aligned} \quad (18)$$

$$H_{\vec{K}} \approx t \sum_{p\sigma} \left(a_{p\sigma}^\dagger b_{p\sigma}^\dagger c_{p\sigma}^\dagger \right) \begin{pmatrix} 0 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 0 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \quad (19)$$

Directly at the $\vec{K}$ -point the eigenvalues are obviously

$$\varepsilon_1(\vec{K}) = t, \quad \varepsilon_2(\vec{K}) = t, \quad \varepsilon_3(\vec{K}) = -2t, \quad (20)$$

the eigenvector corresponding to the flat band reads at the $\vec{K}$ -point:

$$\vec{v}_3(\vec{K}) = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}. \quad (21)$$

Since H is hermitian (/symmetric in this case) there exist a orthonormal basis of eigenvectors. Obviously the subspace orthonormal to $\vec{v}_3(\vec{K})$ corresponds to the two bands that cross. A possible (orthonormal) basis for this subspace is

$$\vec{v}_1(\vec{K}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_2(\vec{K}) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}. \quad (22)$$

Projecting H onto this basis yields

$$\begin{aligned} \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= t - \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} & \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= -t \left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} \\ \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= -t \left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} & \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= t + \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} \\ \Leftrightarrow t\sigma_0 - \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \sigma_z &+ t \left(\left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} \right) \sigma_x \end{aligned} \quad (23)$$

This already has the correct dispersiob

$$\varepsilon_{1,2}(\vec{K} + \vec{p}) \approx \sqrt{3}t|\vec{p}|, \quad (24)$$

but is not yet in the form of the Weyl equation. To transform the equation into the Weyl equation one can rotate the basis inside the degenerate subspace. I take the following approach; define

$$\begin{aligned} \vec{w}_1 &= \alpha \vec{v}_1 + \beta \vec{v}_2 \\ \vec{w}_2 &= \eta \vec{v}_1 + \gamma \vec{v}_2 \end{aligned} \quad (25)$$

Othogonality dictates

$$\begin{aligned} |\alpha|^2 + |\beta|^2 &= 1 \\ |\eta|^2 + |\gamma|^2 &= 1 \\ \alpha^* \eta + \beta^* \gamma &= 0. \end{aligned}$$

Utilizing $\vec{\delta}_2 + \vec{\delta}_3 = \vec{\delta}_1$ I get:

$$\vec{w}_1^T \cdot H_{\vec{K}} \cdot \vec{w}_1 = t + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2|\alpha|^2 - \frac{2\text{Re}(\alpha^* \beta)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{4\text{Re}(\alpha^* \beta)}{\sqrt{3}} \right) \quad (26)$$

$$\vec{w}_2^T \cdot H_{\vec{K}} \cdot \vec{w}_2 = 1 + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2|\eta|^2 - \frac{2\text{Re}(\eta^* \gamma)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{4\text{Re}(\eta^* \gamma)}{\sqrt{3}} \right) \quad (27)$$

$$\vec{w}_1^T \cdot H_{\vec{K}} \cdot \vec{w}_2 = \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2\alpha^* \eta - \frac{\alpha^* \gamma + \beta^* \eta}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{2(\alpha^* \gamma + \beta^* \eta)}{\sqrt{3}} \right) \quad (28)$$

$$\vec{w}_2^T \cdot H_{\vec{K}} \cdot \vec{w}_1 = (\vec{w}_1^T \cdot H_{\vec{K}} \cdot \vec{w}_2)^* \quad (29)$$

The strategy is to solve for values for α, β, γ and η such that the following combinations occur in the matrix elements:

$$p_x = \left(\vec{\delta}_2 + \vec{\delta}_3 \right) \cdot \vec{p} \quad (30)$$

$$p_y = \frac{1}{\sqrt{3}} \left(\vec{\delta}_2 - \vec{\delta}_3 \right) \cdot \vec{p} \quad (31)$$

whil still leading to orthonormal basis vectors. The following parameters are a sensible choice

$$\alpha = -\frac{1}{2}, \quad \beta = \frac{\sqrt{3}}{2}, \quad \eta = \frac{\sqrt{3}}{2}, \quad \gamma = \frac{1}{2}. \quad (32)$$

Thus, the new basis vectors read

$$\vec{w}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad (33)$$

$$\Rightarrow \langle H_{\vec{K}} \rangle_{\vec{w}_1, \vec{w}_2} = t\sigma_0 + \sqrt{3}tp_y\sigma_x + \sqrt{3}tp_x\sigma_z \quad (34)$$

This form was also found in [2], a different one was obtained in [3]. I expect that I would find the latter if I chose different values for α, β, γ and η . I decided to stick with this form to keep the expressions real.

I could eliminate the σ_0 term by introducing chemical potentials but I guess for non-interacting particles it doesn't really matter.

Question: when rotating the orthonormal basis inside the degenerate subspace there is one free parameter (the rotation angle). On the other hand one can probably also think about the surface of a Bloch sphere (the radius is obviously fixed). What is the free parameter in this picture /what is the fixed parameter? It is probably worthwhile to explore this at some point, it might reveal the underlying topology.

0.1.6 Introduction of Strain

I show how to expand expressions prototypical for the k-space Hamiltonian in strain:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}} \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{q} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right), \quad (35)$$

afterwards I expand these expression around $\vec{K}$:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}} \left(1 + \vec{p} \cdot \vec{\delta}_{ij} \right) \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\underline{u} \cdot \vec{\delta}_{ij} \right) \cdot \vec{p} \right) \quad (36)$$

I compared this to the expansion on the triangular lattice, where I used the real space method. The main difference is the order of expansions. As I do the k-space expansion second I find terms of second order in p . For the triangular lattice I did it in opposite order, thereby acquiring terms of second order in $\underline{u}$. This means one can only compare terms of zeroth order and of first order either in $\underline{u}$ or in p . These terms check out. Revise this

Keeping only the relevant terms (relevant to this order at least):

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}} \left(1 + i\vec{p} \cdot \vec{\delta}_{ij} - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right) \quad (37)$$

$$\begin{aligned} \Rightarrow t(-\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} &\approx 2t \cos(\vec{K} \cdot \vec{\delta}_{ij}) - 2t \sin(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{p} \cdot \vec{\delta}_{ij} \right) \\ &\quad - 2\beta t \cos(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \\ &\quad - 2t \sin(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \end{aligned} \quad (38)$$

I group the linear corrections by order in β to find the two matrices:

$$\delta H_1 = -2t\beta \begin{pmatrix} 0 & (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy})/8 \\ (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & 0 & -u_{xx}/2 \\ (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy})/8 & -u_{xx}/2 & 0 \end{pmatrix} \quad (39)$$

$$\delta H_2 = -2t \begin{pmatrix} 0 & \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 \\ \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & 0 & \pi\frac{u_{xx}}{\sqrt{3}} \\ \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 & \pi\frac{u_{xx}}{\sqrt{3}} & 0 \end{pmatrix} \quad (40)$$

The projections are

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -t\beta \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & u_{yy} \end{pmatrix} \quad (41)$$

$$\langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & -u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} (u_{yx}\sigma_y + u_{xx}\sigma_z) \quad (42)$$

$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2}$ can be written as vector potential + scalar potential using:

$$\begin{aligned} u_{xx} &= \frac{1}{2}(u_{xx} - u_{yy}) + \frac{1}{2}(u_{xx} + u_{yy}) \\ u_{yy} &= -\frac{1}{2}(u_{xx} - u_{yy}) + \frac{1}{2}(u_{xx} + u_{yy}). \end{aligned} \quad (43)$$

$$\Rightarrow \langle H_{\vec{K}} + \delta H_1 + \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = (t + \phi)\sigma^0 + \sqrt{3}t(p_y + A_y)\sigma_x + \sqrt{3}t(p_x + A_x)\sigma_z \quad (44)$$

$$\phi = \frac{t}{2} (u_{xx} + u_{yy}) \quad (45)$$

$$A_x = -\frac{t\beta}{2\sqrt{3}} (u_{xx} - u_{yy}) - \frac{2\pi t}{3} u_{xx} \quad (46)$$

$$A_y = -\frac{t\beta}{\sqrt{3}} u_{yx} - \frac{2\pi t}{3} u_{yx} \quad (47)$$

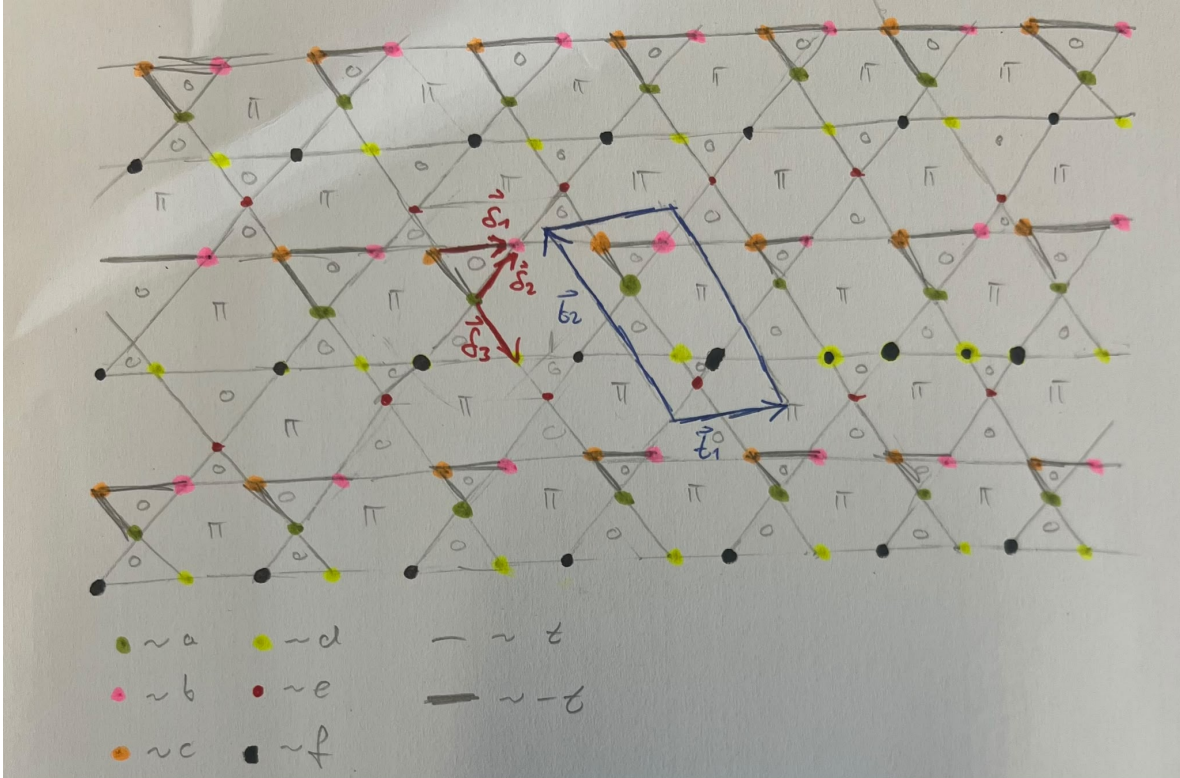
The β -independent part of $\vec{A}$ can be eliminated using a gauge-transformation. This agrees with the results here [3].

Its Fleißarbeit to do the same for $\vec{K}'$.

0.2 π -flux State

0.2.1 Real Space

This lattice is also a triangular Bravais-lattice, only now with unit cell doubling (in the direction of $\vec{t}_2$), hence the unit cell has 6 sites. The Wigner-Seitz cell is a stretched hexagon. In the cartoon a different unit cell is shown.



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \quad (48)$$

The new primitive translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \quad (49)$$

The real space (effective) tight-binding hamiltonian is

$$H = t \sum_{i\sigma} \left(-a_{i\sigma}^\dagger c_{i\sigma} - c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + a_{i\sigma}^\dagger d_{i\sigma} + d_{i\sigma}^\dagger e_{i\sigma} + e_{i\sigma}^\dagger f_{i\sigma} + f_{i\sigma}^\dagger d_{i\sigma} \right. \\ \left. + b_{i\sigma} c_{i+t_1;\sigma} + b_{i\sigma} e_{i+t_1+t_2;\sigma} + f_{i\sigma} d_{i+t_1;\sigma} + f_{i\sigma} c_{i-t_2;\sigma} + \text{h.c.} \right) \quad (50)$$

0.2.2 Reciprocal Space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \quad (51)$$

After a Fourier transform the k -space Hamiltonian reads

$$H = t \sum_{q,\sigma} \Psi_{q\sigma}^\dagger \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i \sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{iq_3} & -2i \sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2 \cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2 \cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \quad (52)$$

where

$$\Psi_{q\sigma} = (a_{q\sigma} \quad b_{q\sigma} \quad c_{q\sigma} \quad e_{q\sigma} \quad f_{q\sigma} \quad d_{q\sigma}). \quad (53)$$

If I write this down somewhere serious, I should just label the sites differently.

I used the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$.

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